

Parth Joshi

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EXPERIENCE

Project Engineering Intern

May 2025 – Present

Bruce Power (Seconded by TetraTech)

Tiverton, ON

- Built a Power BI dashboard providing real-time visibility into engineering requests and providing the key **KPIs** to the relevant stakeholders in clear way
- Connected **Power BI** and Power Apps to an **Azure SQL** database sourced from Oracle, pulling structured data to support real-time reporting and application tables
- Engineered custom JavaScript logic to solve a **many-to-many (n:n)** relationship in Power Apps, enabling accurate record linking across related work packages and requests
- Automated data updates and request routing using Power Automate, reducing manual data entry and cutting tracking errors/delays by an estimated **15%**

PROJECTS

Neural Network Racing Simulator | C, OpenGL, Python | [GitHub](#)

Mar 2026

- Built a real-time physics simulation engine in C with custom memory management, collision detection, and ray-casting sensor systems for reinforcement learning agents
- Trained the driving policy end-to-end with a **REINFORCE** - style policy-gradient algorithm, Gaussian action sampling with sigma decay, reward shaping, and gradient clipping over **30,000+ episodes**
- Accelerated collision and sensor queries with a custom quadtree spatial index over track geometry, paired with real-time OpenGL rendering for visualization
- Developed neural network training from scratch (forward pass, backpropagation, gradient descent) without libraries

Maple Sap Flow Prediction | Python, FastAPI, HTML/CSS/JavaScript | [GitHub](#)

Nov 2025

- Built a full-stack forecasting application that analyzed **500K+ geospatial** observations to predict maple sap-flow windows, winning BramHacks 2025 and a **\$400 prize**
- Developed a Python data pipeline to process and transform **20+ years** of climate and geospatial data from NASA Earth Engine into model-ready features
- Implemented feature engineering, normalization, and trend-analysis methods to capture seasonal weather patterns associated with sap flow

AI Appointment Follow-Up System | Python, FastAPI, SQL, React, WebSockets | [GitHub](#)

Jan 2026

- Designed and built an AI-powered outbound calling system automating appointment confirmations and rescheduling using real-time voice interactions
- Implemented WebSocket communication and real-time pipelines for structured scheduling data
- Developed full-stack system with SQL-backed data models for managing patients, appointments, and outcomes

Chess Engine | Python, Game Tree Search | [GitHub](#)

Jan 2026

- Developed a high-performance chess engine using alpha-beta pruning, transposition tables, and advanced search optimizations (**~2000 Elo**)
- Optimized computation using efficient memory reuse and incremental state updates, achieving **10–25K nodes/sec** search throughput
- Implemented full rule engine supporting complex game logic and real-time decision systems

EDUCATION

Western University

London, ON

Honors Spec in Computer Science + Math

Sept 2024 – June 2028

- Cumulative GPA: 3.72
- Relevant Coursework: Operating Systems, Data Structures and Algorithms, Discrete Math, Numerical Computing, Statistics

SKILLS & INTERESTS

Languages: C, Python, Java, JavaScript, SQL

Software: Data Structures, Algorithms, Object-Oriented Design

Tools: Git, Linux, OpenGL, REST APIs, Docker, Unix, Azure

Interests: Performance Optimization, Machine Learning, Chess, AI/LLMs